

Reliability

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Classical Test Theory

$$X = T + E$$



Classical Test Theory

Random error

- Random in nature
- Will increase and decrease a person's score by exactly the same amount with infinite testing
- Cancels itself out
- Lowers reliability of a test

Systematic error

- Occurs when source of error always increase or decrease a true score
- Does not lower reliability of a test since the test is reliably inaccurate by the same amount each time



Reliability in Survey

If researchers are trying to determine how an individual feels about age discrimination in the workplace, they might ask this question:

1a. How much do you agree or disagree with the U.S. Supreme Court's 2013 ruling on *Madigan v. Levin*?

1. Strongly agree
2. Somewhat agree
3. Somewhat disagree
4. Strongly disagree



Reliability in Survey

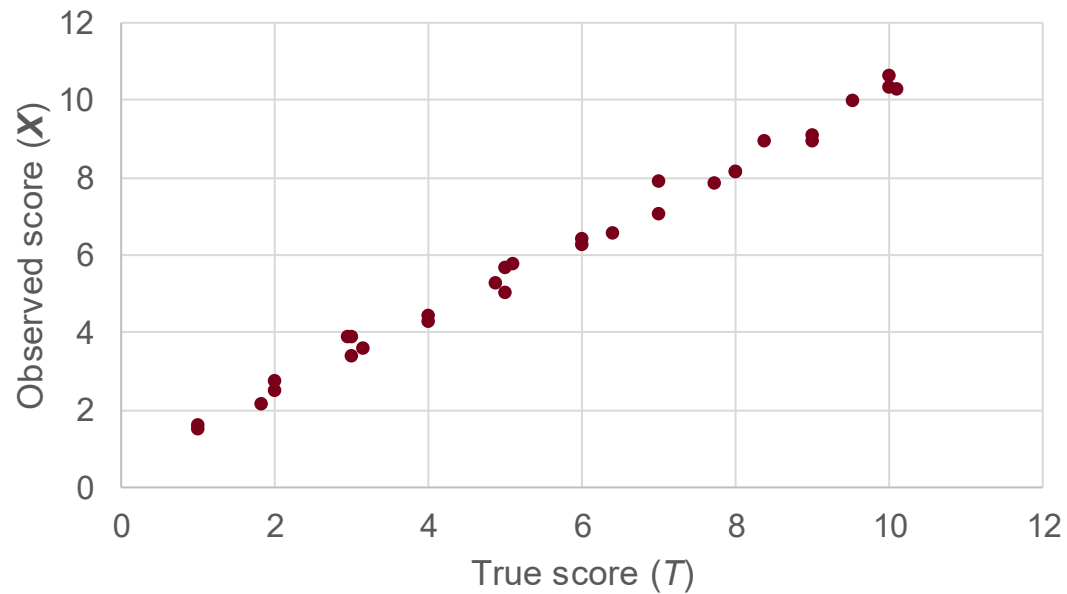
1b. Please tell us how much you agree or disagree with the statement: An employer has the right to terminate employment based on an individual's age?

1. Strongly agree
2. Somewhat agree
3. Somewhat disagree
4. Strongly disagree



Estimating reliability

$$X = T + E$$



Estimating Reliability

- The theoretical reliability coefficient is not practical; we do not know each person's true score.
- Nevertheless, we can estimate the theoretical coefficient with the sample correlation between scores on two parallel tests.



Reliability

- Reliability refers to the precision of the measurements, or the consistency of scores produced by a test.
- Reliability is not a property of the test, but that of scores.
- If scores are consistent, the measurement is reliable.
- Scores could be inconsistent in varied ways – reliability should be assessed accordingly



Estimating reliability

	XY		XY
+1.0 → perfect positive linear relationship:	↑↑	or	↓↓
0.8 to 1.0 → strong positive linear relationship:	↑↑	or	↓↓
0.4 to 0.7 → moderate positive linear relationship:	↑↑	or	↓↓
0.1 to 0.3 → weak positive linear relationship:	↑↑	or	↓↓
0 → no linear relationship	↑-	or	↓-
-0.1 to -0.3 → weak negative linear relationship:	↑↓	or	↓↑
-0.4 to -0.7 → moderate negative linear relationship:	↑↓	or	↓↑
-0.8 to -1.0 → strong positive linear relationship:	↑↓	or	↓↑
-1.0 → perfect negative linear relationship:	↑↓	or	↓↑



Types of reliability

- Test–retest reliability
 - consistency of scores over time
- Alternate forms reliability
 - consistency of scores over multiple test forms
- Internal consistency
 - consistency of scores across test items
- Interrater agreement
 - consistency of scores across raters



Test–retest reliability

- Test–retest reliability is concerned with the consistency, or stability, of test scores across different time points.
- Test scores can change across time because of
 - 1) lack of reliability of the measure being used,
 - 2) changes in respondents' true levels of the construct being measured, or
 - 3) a combination of these.



- How many times have you attended religious services in your entire life? _____ (T1/T2)



- In the previous month, how many times have you attended religious services? (T1/T2)
 - 0
 - 1–2
 - 3–4
 - 5 or more



Recommended Values for Coefficients of Stability

- Crocker and Algina (1986, p. 133) state that “well-constructed” noncognitive scales can yield scores with coefficients in the .80s, and scores from aptitude tests can have values into the .90s.
- Coefficients of stability would likely be higher if the time interval had been shorter, and lower if the time interval had been longer.



Internal Consistency

- I consider myself to be
 - Not very religious
 - Somewhat religious
 - Very religious
- I attend religious services
 - Not very often
 - Somewhat often
 - Very often
- I read religious texts
 - Not very often
 - Somewhat often
 - Very often



Internal Consistency

- Internal Consistency concerns about the source of error from the items of the test.
- An early approach to estimating the internal reliability of a test was to split the test into two halves and correlate the resulting two scores, followed by an adjustment.
- Coefficient alpha is one of the more popular measures for internal consistency; equivalent to the average of all possible split halves of a test



Internal Consistency

- Coefficient alpha is only suitable under some strong assumptions, which are hard to met in practice
- Coefficient omega can often overcome the deficiencies of coefficient alpha



Misconceptions about Coefficient Alpha

- α is one of the most popular psychometric indices
- There are several common misconceptions about its interpretation
 - Let's talk about what α actually is and what it is not

What α is; What α is not

What α is	What α is not
1. Index of internal consistency	1. Indicator of dimensionality
2. Mean of all possible split-half reliability coefficients	2. Lower-bound of reliability
3. Estimate of reliability in certain circumstances	



Recommended Values for Internal Consistency Indices

- Raykov and Marcoulides (2011) recommend values of at least .80.
- Nunnally's (1978) provides the following guidelines:
 - .70 for scales in the initial stages of development;
 - .80 for “basic research scales,” and
 - .90 as minimum for use in clinical settings.
- Kaplan and Sarcuzzo (2001) suggest values of up to .95 for clinical settings.



How to improve reliability?

- Clarity: Clearly written surveys that avoid ambiguous language in questions and response options are much more reliable sources of data than less carefully constructed instruments
- Brevity: Not only the language, but the mere number of possible response options can strongly influence how an individual is going to respond
- Length: reliability is usually increased as test length increases.



- 1a. How many times have you attended religious services in the past year? _____
- 1b. How many times have you attended religious services in the past year?
 - 0
 - 1–3
 - 4–6
 - 7–10
 - 10–12
 - 13 or more
- 1c. How many times have you attended religious services in the past year?
 - 0
 - 1–3
 - 4–5
 - 6 or more

